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Camille Robichon

PhD Candidate in Food Science

Research Profile

Food scientist and formulation chemist focusing on synergistic antioxidant interactions in food matrices. Strong expertise in lipid model systems, emulsions, oxidative stability monitoring, and liposome formulation, using HPLC, NMR and UV-Vis analyses. International experience in France, Scotland and Japan. Currently pursuing a PhD in Food Science and seeking research and development project-driven positions in lipid oxidation, antioxidant interactions, bioactive delivery systems, functional compounds...

Education

2023–Present **PhD in Food Science**, *University of Montpellier*, Montpellier, France
Thesis: *New Vitamin Formulation Pathways to Increase Antioxidant Efficacy in Food Matrices.*

2020–2023 **Chemical Engineering Degree, Formulation Chemistry**, *École Nationale Supérieure de Chimie de Lille (ENSCL)*, Lille, France
Specialisation in formulation chemistry; research project on the synthesis of flame retardants for polylactide (PLA).

Sep 2022–Mar 2023 **Exchange Semester**, *Doshisha University*, Kyoto, Japan
Project: Transdermal administration of curcumin using an oil-in-water microemulsion based on isopropyl myristate; spectrophotometric studies and Franz cell experiments.

Research Experience

Nov 2023–Present **PhD Candidate in Food Science**, *CIRAD, UMR QualiSud*, Montpellier, France
New Vitamin Formulation Pathways to Increase Antioxidant Efficacy in Food Matrices.
- Investigate synergistic antioxidant interactions between vitamin E (tocopherols) and natural antioxidants in oil-in-water emulsions.
- Develop lipid model systems and nanoemulsions using a Microfluidizer.
- Formulate and characterise liposomes (DLS, HPLC).
- Study oxidative stability of food matrices via hydroperoxide monitoring (NMR, UV-Vis) and antioxidant consumption (HPLC).
- Implement *in vitro* digestion protocols and evaluate antioxidant bioaccessibility.

2023 (6 months) **End-of-studies Research Project**, *BioWooEB Research Group, CIRAD*, Montpellier, France
- Optimised a method for extracting proteins from dry rubber.
- Performed SDS-PAGE electrophoresis analyses.
- Developed an infrared (IR)-based method for the characterisation of rubber proteins.

2022 (2 months) **Research Internship**, *ADS Research Group, University of St Andrews*, St Andrews, Scotland, UK
- Synthesised pyrazolone derivatives and carried out kinetic resolution by isothiourea catalysis.
- Purified compounds using column chromatography.
- Performed HPLC, NMR and IR analyses for structural and purity assessment.

Publications

2025 Robichon, C., Durand, E., Gamaethiralalage, J.G., Bohuon, P., Baréa, B., Barouh, N., Courtois, F., Fine, F., De Smet, L., Villeneuve, P. *Decoding tocopherol–polyphenol interactions in oil-in-water emulsions through combined WIM-CAT and CV assays*. *Current Research in Food Science* (submitted).

2025 Robichon, C., Villeneuve, P., Bohuon, P., Baréa, B., Barouh, N., Courtois, F., Fine, F., Durand, E. *Unveiling synergistic antioxidant combinations for α-tocopherol in emulsions: a spectrophotometric-mathematical approach*. *Current Research in Food Science*, 2025.

2024 Smith, A.D., Westwood, M.T., Farah, A.O., Wise, H.B., Sinfield, M., Robichon, C., Prindl, M.I., Cordes, D.B., Cheong, P.H.-Y. *Isothiourea-catalysed acylative kinetic resolution of tertiary pyrazolone alcohols*. *Angewandte Chemie International Edition*, 2024.

Scientific Communications

Oral EuroFedLipid 2025, Leipzig, Germany. *Design of a spectrophotometric assay coupled with mathematical modelling to identify optimal antioxidant synergies with tocopherols in O/W emulsions*.

Oral AOCS 2025, Portland, USA. *Development of a spectrophotometric assay with mathematical modelling to reveal optimal antioxidant combinations with tocopherols in O/W emulsions*.

Poster EuroFedSymposium 2024. *A rapid and tunable screening method for exploring synergistic antioxidant interactions with α-tocopherol in O/W emulsions*.

Software & Code

Model comparison MATLAB framework comparing kinetic models (Weibull, Gompertz, Logistic, Exponential) for oxidation kinetics in emulsion systems.
`github.com/camillerobichon30/Antioxidant-Kinetics-Model-Comparison`

Antioxidant modelling MATLAB scripts for kinetic modelling of lipid oxidation using Weibull-based functions to evaluate antioxidant efficiency and interaction effects (synergy / additivity / antagonism).
`github.com/camillerobichon30/Weibull-Modelling-for-Antioxidant-Interaction-Eff`

Technical Skills

Experimental Lipid model systems; oil-in-water emulsions; lipid oxidation; antioxidant systems; liposomes; nanoemulsions; Microfluidizer processing; *in vitro* digestion; transdermal delivery systems (Franz cell).

Analytical HPLC; NMR; UV-Vis spectroscopy; IR spectroscopy; DLS; SDS-PAGE.

Computing Tools Python; MATLAB (Curve Fitting Toolbox); Microsoft Office 365; VBA.

Research Interests

Lipid oxidation; antioxidant synergies; food emulsions; delivery systems for bioactive compounds; bioaccessibility and digestion of lipids and antioxidants.

Languages

French	Native
English	Advanced (TOEIC 855)
Spanish	Intermediate
Japanese	Beginner; currently learning

Interests

Reading (detective novels); cycling; walking and hiking; concerts; learning Japanese language and culture.